Notes on rCCA Reproduction Results

When reviewing the attached notebook (`rcca_combinedwith_s2s_firstresults.ipynb`) and comparing my output to the results from the original paper, you will notice that my results are scientifically consistent, though the exact numerical skill scores differ slightly.

Here is a visual comparison of the extracted cluster centers:

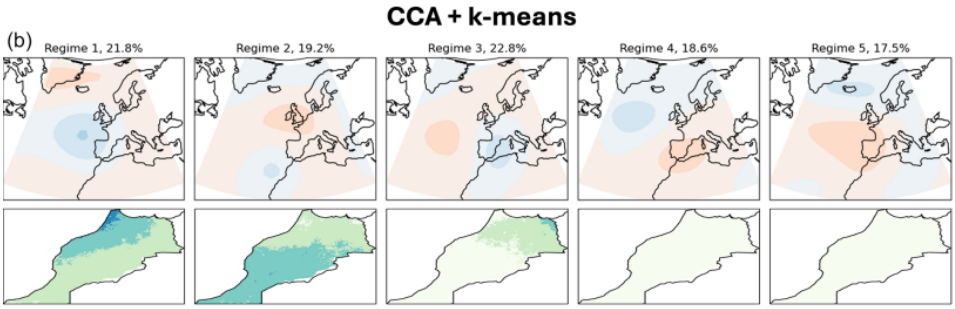

Figure 1: Cluster centers from the original paper.

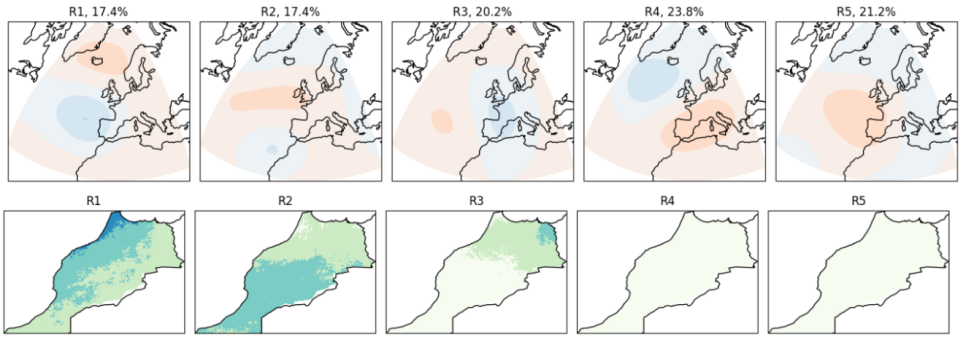

Figure 2: My reproduced cluster centers.

Below is an explanation of the challenge I encountered during the reproduction process and how I resolved it:

1. The Missing Data Problem

The main issue I faced was that the original CCA + K-Means pipeline relied on a specific dataset that was missing from our repository: `total_pr_admin_areas.nc`.

According to Section 2.2.2 of the paper:

> *We therefore aggregated precipitation at the district level over Morocco and implemented a ridge regularization parameter for the geopotential height field.*

Because this pre-processed file containing precipitation data explicitly averaged over exact Moroccan administrative districts was not available, I could not feed the exact same "second view" into the Regularized Canonical Correlation Analysis (rCCA).

2. My Workaround Solution

To allow the methodology to run and produce meaningful results without requiring new dependencies or external shapefiles, I implemented a **spatial chunking approximation**.

Instead of using the missing administrative regions, my code derives 10 arbitrary, contiguous spatial chunks directly from the available `CHIRPS_pr_total_EW.nc` precipitation field. I flattened the spatial dimension, divided it into 10 equal segments, and took the mean of each chunk.

I then applied the identical Box-Cox transformation to these 10 chunks and fed them into the rCCA algorithm.